19 May 1976
Superseding
SCL-4296D
8 December 1969

MILITARY SPECIFICATION

MAST AB-864( )/G and MAST EXTENSION KIT MK-1624( )/G

This specification is approved for use by the Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for a guyed, lightweight 50-foot mast and for a 20-foot mast extension kit, transportable in 3/4-ton and 1-1/4-ton shelter assemblages. The mast is for support of lightweight antennas or other equipment where erection speed and low manpower requirements are prime considerations. This mast is extendable to a 70-foot height using the extension kit.

1.2 Description.

1.2.1 The mast consists of a high-speed launching device with eight mast sections to extend the mast by manually inserting each section in the launcher. A lifting device is provided in the launcher to vertically elevate each section to full height. The mast is rugged and compact and capable of erection in less than 30 minutes with a 3 man crew. All parts are erected under blackout conditions. Other markings and color codings are used to facilitate daytime assembly. Provisions are made for 360 degree aximuth rotation of the mast with means for locking at any bearing. Azimuth markings are provided in the base assembly at five degree increments and numerals at each fifteen-degree point. All necessary hardware, tools and facilities for storage and transit are provided. The mast and its accessories are capable of being erected and dismantled by soldiers wearing arctic clothing and mittens. The weight of the basic 50-foot mast is approximately 225 pounds.

1.2.2 The mast extension kit consists of a four (4) mast section pack and four (4) guy extensions and is approximately 60 pounds.

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2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	Strapping, Steel and Seals
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated - Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-76	Tape, Pressure Sensitive Adhesive Paper, (For Carton Sealing)

MILITARY

MIL-P-116	Preservation, Methods of
MIL-P-11268	Parts, Materials and Processes Used In Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-C-13924	Coating, Oxide, Black, for Ferrous Metals
MIL-F-14072	Finishes for Ground Signal Equipment

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

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MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

DL-SM-B-728684	Data and Drawing List for Mast AB-864( )/G
DL-SM-B-728712	Data and Drawing List for Mast Extension Kit MK-1624( )/G

3. REQUIREMENTS

3.1 First Article.- When specified in the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.

3.2 Construction.- The mast and mast extension kit shall be constructed in accordance with the requirements of this specification and the drawings listed on DL-SM-B-728684 and DL-SM-B-728712, respectively.

3.2.1 Selection and use of parts, materials and processes.- If the contractor proposes the use of a part, material or process other than that specified on drawings, specifications or contract, the selection and approval procedures of MIL-P-11268 are a requirement. The above includes the proposed use of "or equal" parts, materials and processes. (See 6.21)

3.3 Cleaning.- (See 3.19 and 4.4)

3.3.1 Parts.- After fabrication, parts shall be cleaned in accordance with requirement 9 of MIL-STD-454 except as specified in an applicable document.

3.3.2 Units.- After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux and other foreign material. In addition, such cleaning shall also be performed before final assembly of the equipment.

3.4 Finish.- The mast and mast extension kit shall be finished in accordance with MIL-F-14072 and the drawings. Exterior stainless steel shall be black oxide coated conforming to MIL-C-13924. (See

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3.19 and 4.4) The final paint film shall be olive drab color No. 24087 per FED-STD-595. (See 6.5)

3.5 Marking.- Marking shall conform to MIL-M-13231 and the drawings. (See 3.19 and 4.4)

3.5.1 Serial numbers.- Each mast shall be serial numbered. Numerical sequence shall be supplied by the contracting officer.

3.6 Riveting.- Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces. (See 3.19)

3.7 Securing of parts.- Brackets, lugs, flanges, inserts, bolts and other mounting arrangements shall be suitably tight to retain items securely when subjected to specified bounce and shock tests. (See 3.14 and 3.19)

3.8 Tropicalization.-

3.8.1 Treatment of Materials.- Fungus-inert materials shall be used where practicable. These materials shall be in accordance with requirement 4 of MIL-STD-454. When materials are required to be treated, they shall be cleaned in accordance with requirement 9 of MIL-STD-454 and then shall be treated in accordance with 3.8.2 thru 3.8.5 and as follows:

Fibrous material, organic (See MIL-P-11268)
Cork (See MIL-P-11268)
Felt, hair or wool (See MIL-P-11268)
Thermosetting plastic materials using cotton, linen or wood-flour as a filler or base (See MIL-P-11268)

3.8.2 Treating compounds.- Treating compounds containing a mercury bearing fungicide shall not be used. The contractor shall determine that the treating compound is compatible with the material or surface to be treated. Selection of treating compounds shall be such that there shall be no increase in flammability of the treated material.

3.8.3 Toxicity.- Treating compounds shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

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3.8.4 Flexibility.- Treatment shall not affect the flexibility of the treated material to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.8.5 Statement of treatment.- The statement of treatment shall indicate whether or not nutrient materials have been employed. If nutrient materials have been used, it shall also describe in detail the materials to be treated and the treating compounds and processes that the contractor proposes to use. (See 6.2h)

3.8.6 Fungus.- The equipment shall show no evidence of viable fungus or corrosion when tested per 4.18. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish or metal surface.

3.9 Launcher.-

3.9.1 Brake.- Launcher brake shall be capable of supporting an axial mast load of 1000 pounds minimum without slippage. (See 4.6.1)

3.9.2 Bending.- The guyed launcher shall be capable of sustaining a bending moment of 50,000 inch pounds minimum applied to the mast as specified in 4.6.2 without failure or permanent deformation greater than 3/32 inch. The launcher's function shall not be affected after load is removed.

3.10 Winch.-

3.10.1 Load.- The winch assembly shall be capable of raising a 500-pound load minimum without damage to the assembly. (See 4.7.1)

3.10.2 Brake.- The winch brake shall hold a 500-pound minimum axial mast load without slipping. (See 4.7.2)

3.10.3 Torque.- The handle torque for raising a 500-pound minimum axial load shall not exceed 30-foot pounds. (See 4.7.3)

3.11 Mast section.- (See 4.8)

3.11.1 Bending.- The mast section shall not fail prior to attaining a bending moment of 50,000 inch pounds minimum with loads applied in accordance with 4.8.1. The mast section index assembly shall

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withstand the load of 50,000 inch pounds minimum when the index assembly is located on the neutral axis as well as in the compression/ tension axis.

3.11.2 Bonding.- The metal bonding technique shall be such that the tubing fails in test prior to the bond when loaded in accordance with 4.8.2.

3.11.3 Indexing.- The total torsional displacement of the mast sections with mast cap shall not exceed ± 1.5 degrees per eight (8) sections when subject to a torsional load of 100 inch pounds minimum in both clockwise and counter-clockwise direction. (See 4. 8.3)

3.12 Guys.-

3.12.1 Guy assemblies and guy extension cables.- These items shall withstand 2000 pound load minimum without failure, slippage of cable through crimped sleeve, or permanent deformation of cable hook. (See 4.9.1)

3.12.2 Tensioner.-

3.12.2.1 Guy and tensioner assembly.- The tensioners shall withstand 2000 pound load minimum without failure or slippage of cable when tested with the safety latch disengaged (locked open). Cable may show signs of surface abrasion and damage, however, the damage shall not prevent reuse of the guy. (See 4.9.2.1)

3.12.2.2 Tensioner Load.- The tensioner shall be capable of withstanding a 4000 pound minimum tensile load applied between the tensioner drum and the guy extension attachment without failure. (See 4.9.2.2)

3.12.2.3 Tensioner locking handle.- The locking handle shall be capable of sustaining a load of 40 pounds minimum without permanent deformation. The 40 pound load shall be applied at the free end so that a torque is developed in the unlocking direction at the base of the handle where it attached to the load transfer element. (See 4.9.2.3)

3.13 Service conditions.- The equipment shall sustain no corrosion, pitting, hardening, cracking, softening or tackiness of rubber or plastic materials, decomposition of cloth, bonding failure, binding of moving parts, deformation, or other deterioration of materials or operation when subjected to the following service conditions.

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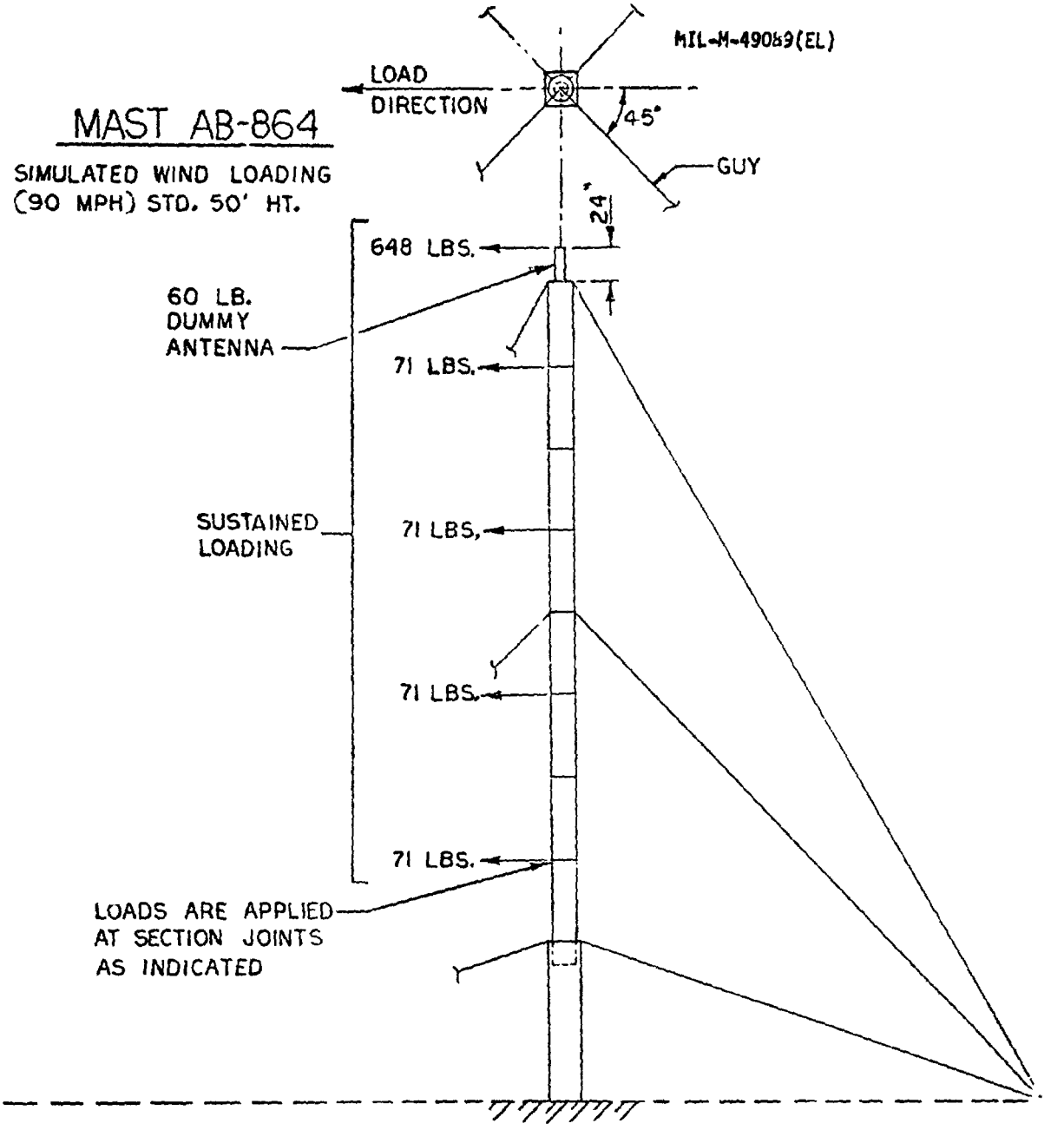
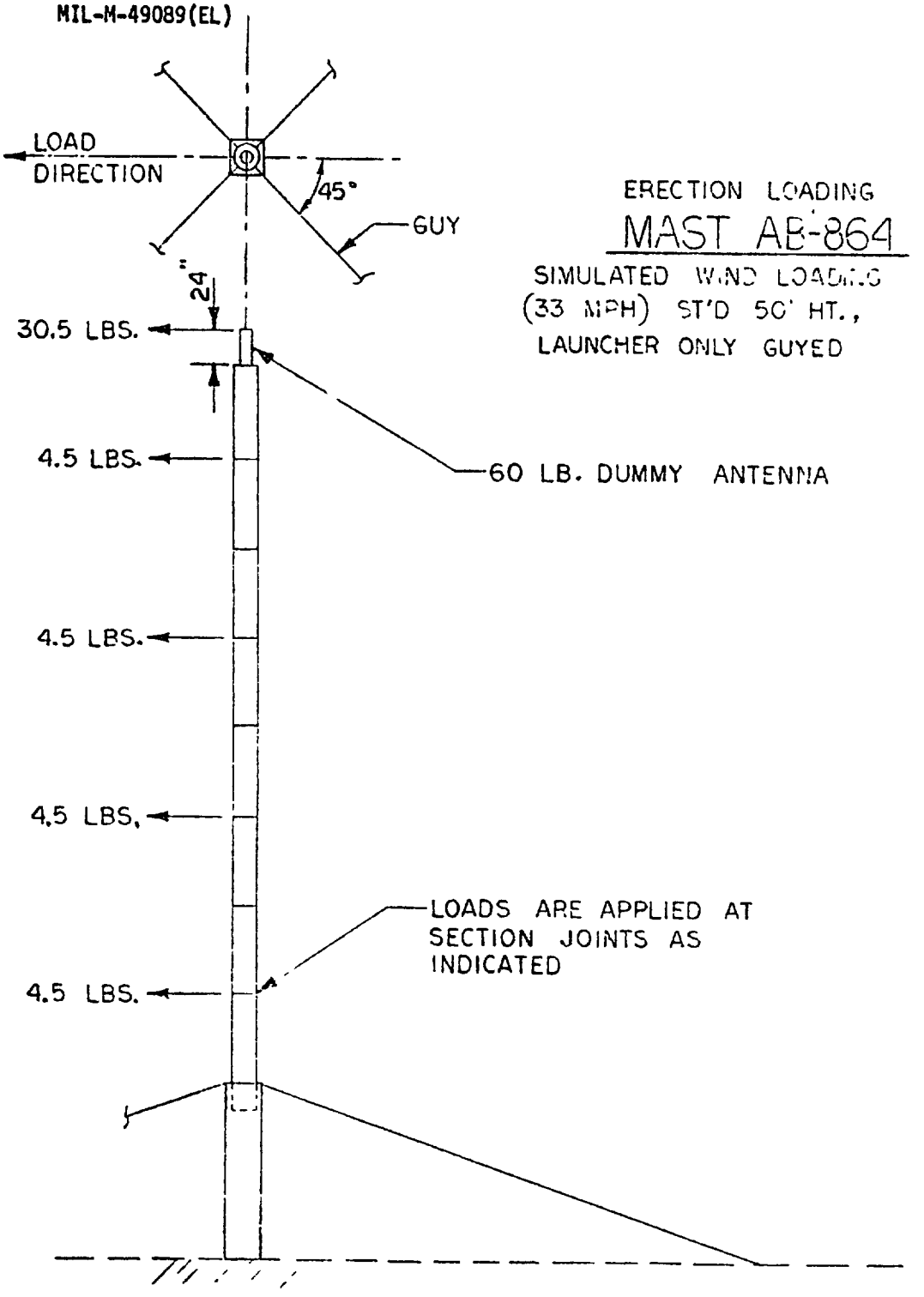


FIGURE 1

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ERECTION LOADING
MAST AB-864
SIMULATED WIND LOADING
(33 MPH) ST'D 50' HT.,
LAUNCHER ONLY GUYED

FIGURE 2

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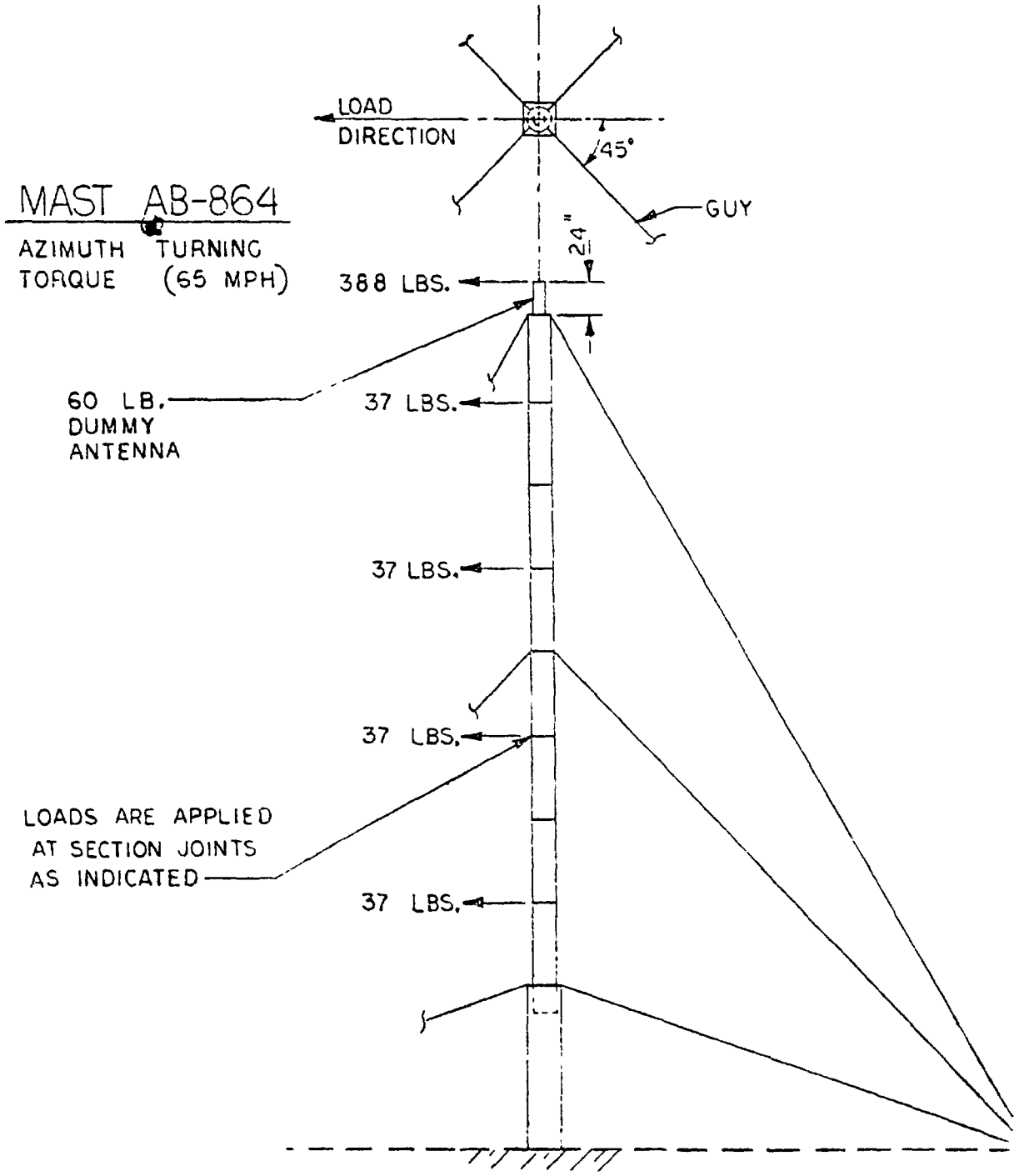


FIGURE 3

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3.13.1 Temperature.- (See 4.13)

a. Operating.- Ambient temperature of -70°F. to +160°F.

b. Non-operating.- Temperature in the range of -70°F. to +160°F.

3.13.2 Humidity.- Exposure to warm, highly humid (tropical) atmosphere. (See 4.14)

3.13.3 Salt fog.- Exposure to salt (coastal) atmosphere. (See 4.17)

3.13.4 Dust.- Exposure to sand (desert) or dust atmosphere. (See 4.16)

3.13.5 Altitude.- Exposure to reduced pressure (air shipment or high ground operation). (See 4.15)

3.14 Bounce and shock.-

3.14.1 Bounce, loose cargo.- The equipment shall sustain no physical damage except minor surface abrasion when tested per 4.11. Degradation in performance in not permissible.

3.14.2 Shock, drop.- The equipment shall sustain no structural damage when tested per 4.12. Denting of the point of contact is permissible. Weld cracks, connector failures, deformation of any member, or degradation in performance is not permissible.

3.15 Loading, mast.- (See 4.10.2)

3.15.1 Sustained loading 50-foot mast.- With anchors located, using anchor rope, 35 feet from the center and fully guyed, the mast shall withstand simulated 90 mph wind loads applied as per Figure 1 and Paragraph 4.10.2.1 with no failure or permanent deformation to any part.

3.15.2 Erection Load.- The mast, when guyed at the launcher only, shall withstand the simulated 33 mph wind loads applied per Figure 2 and Paragraph 4.10.2.2 with no failure or permanent deformation to any part.

3.16 Mast operation.- (See 4.19) A 60-pound minimum dummy antenna will be raised a minimum of two (2) mast sections with results as follows:

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a. All mechanical elements such as the brake, gears, bearings and sliding contact surfaces shall be free of binding and chafing.

b. The mast and winch brakes shall function properly while supporting the load.

c. The winch shall turn freely and the crank handle torque shall not exceed 10 foot pounds.

3.17 Rotational resistance.- With the mast at the 50-foot height, fully guyed in the operational mode with a 60-pound minimum dummy antenna at the top and with simulated winds applied per Figure 3, the mast shall be rotatable by a maximum torque of 70 foot pounds. The mast brake shall be in the "locked open" position. (See 4.10.1)

3.18 Interchangeability.- All like parts shall be physically and functionally interchangeable, without modification of the parts. Individual parts shall not be hand picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristics, etc. (See 4.20)

3.19 Workmanship.- The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and with the following paragraphs herein. (See 4.21)

3.2 Construction
3.3 Cleaning
3.4 Finish
3.5 Marking
3.6 Riveting
3.7 Securing of parts

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services to prescribed requirements.

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4.2 Classification of inspection.- The inspection requirements specified herein are classified as follows:

a. First Article inspection (does not include preparation for delivery). (See 4.3)

b. Inspections covered by subsidiary documents. (See 4.4)

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.22)

4.3 First Article.- Unless otherwise specified on the contract, the first article inspection shall be performed by the contractor. (See 3.1)

4.3.1 First Article units.- The contractor shall furnish three (3) First Article units of the complete Mast AB-864( )/G and three (3) First Article units of the Mast Extension Kit MK-1624( )/G.

4.3.2 First Article inspection.- The First Article inspections shall consist of inspections specified in Table I, and shall be performed in the order specified therein.

4.3.3 First Article data.- The First Article test plan and test report shall be as required in the contract.

TABLE I - FIRST ARTICLE INSPECTION

Inspection	REQ'T PARA	INSP PARA	ORDER OF TESTING					
			MAST			EXTEN. KIT		
			#1	#2	#3	#1	#2	#3
Subsidiary documents		4.4	1	1	1	1	1	1
Visual and Mechanical	3.19	4.21	2	2	2	2	2	2

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TABLE I - FIRST ARTICLE INSPECTION
Cont'd

Inspection	REQ'T PARA	INSP PARA	ORDER OF TESTING					
			MAST			EXTEN. KIT		
			#1	#2	#3	#1	#2	#3
Winch	3.10	4.7	3,12	3	3	*	*	*
Launcher	3.9	4.6	4	4	4	*	*	*
Interchangeability	3.18	4.20	5	5	5	3	3	3
Tensioner	3.12.2	4.9.2	*	6	6	4	*	*
Guy Assemblies and Guy Extension Cables	3.12.1	4.9.1	*	7	7	*	4	*
Mast section	3.11	4.8	6	*	*	6	*	*
Mast:		4.10						
Rotational Resistance	3.17	4.10.1	7	*	*	*	*	*
Loading	3.15	4.10.2	*	8	*	*	*	*
Bounce, loose cargo	3.14.1	4.11	8	*	*	7	*	*
Shock, drop	3.14.2	4.12	9	*	*	8	*	*
Mast loading	3.15	4.10.2	10	*	*	*	*	*
Altitude	3.13.5	4.15	11	*	*	*	*	*
Dust	3.13.4	4.16	13	*	*	*	*	*
Fungus	3.8.6	4.18	14	*	*	9	*	*
Temperature	3.13.1	4.13	*	9	*	*	5	*
Humidity	3.13.2	4.14	*	10	*	*	6	*
Salt Fog	3.13.3	4.17	*	*	8	*	*	4
Mast Operation	3.16	4.19	15	11	*	*	*	*

* Omit

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection during production and before preparation for delivery.

3.3 Cleaning
3.4 Finish
3.5 Marking

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 thru 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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4.5.1 Group A Inspection.- Each unit produced shall be inspected for conformance to all the inspection requirements of Table II. Each lot shall then be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspectional levels and the AQL's indicated in Table II. Group A inspection shall be performed in any order which is satisfactory to the Government except that visual and mechanical inspection shall be first.

TABLE II - GROUP A INSPECTION

INSPECTION	REQ'T PARA	INSP PARA	AQL% MAJOR	AQL% MINOR
Visual and Mechanical	3.19	4.21	1%	4%
Winch	3.10	4.7	1%	*
Launcher Brake	3.9.1	4.6.1	1%	*

* All defects are major.

4.5.2 Group B Inspection.- This inspection, including sampling, shall conform to Table III and to the special procedures for small-sample inspection of MIL-STD-105. Inspection level shall be S-4 for normal inspection and S-3 for reduced inspection.

TABLE III - GROUP B INSPECTION

INSPECTION	REQ'T PARA	INSP PARA	AQL%
Interchangeability	3.18	4.20	6.5%

4.5.3 Group C Inspection.- This inspection shall be as listed in Table IV and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspections. Sample units shall be selected in accordance with 4.5.3.1.

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4.5.3.1 Sample for Group C Inspection.- Sample units shall be selected at random for each Group C inspection listed in Table IV in accordance with the following:

4.5.3.1.1 Subgroup I.- One unit from the first 40 production units or fraction thereof and one unit from each successive 40 production units or fraction thereof shall be subjected to the test of Subgroup I.

4.5.3.1.2 Subgroup II and III.- One unit from the first 200 production units or fraction thereof, and one unit from each successive 200 or fraction thereof, shall be subjected to the tests of Subgroup II and III. Subgroup III tests shall be performed in the order given in Table IV.

4.5.3.1.3 Subgroup IV.- Three units from the first 40 production units or fraction thereof shall be subjected to the tests of Subgroup as follows:

a. One unit shall be subjected to the altitude, dust and fungus tests, in that order.

b. A second unit shall be subjected to the temperature and humidity tests, in that order.

c. A third unit shall be subjected to the salt fog test.

4.5.3.2 Group C Failures.- Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on contract, provided that they are re-subjected to and pass Group A inspection after repair of all visible damage.

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TABLE IV - GROUP C INSPECTION

INSPECTION	REQ'T PARA	INSP PARA
Subgroup I	3.12.2	
Tensioner	3.12.2	4.9.2
Guy Assembly and Guy Tension Cables	3.12.1	4.9.1
Subgroup II		
Winch	3.10	4.7
Launcher Bending	3.9.2	4.6.2
Mast Section	3.11	4.8
Subgroup III		
Rotational Resistance	3.17	4.10.1
Mast Loading	3.15	4.10.2
Bounce, loose cargo	3.14.1	4.11
Shock, drop	3.14.2	4.12
Mast, Operation	3.16	4.19
Subgroup IV		
Altitude	3.13.5	4.15
Dust	3.13.4	4.16
Fungus	3.8.6	4.18
Temperature	3.13.1	4.13
Humidity	3.13.2	4.14
Salt Fog	3.13.3	4.17
Mast, Operation	3.16	4.19

4.6 Launcher Tests.-

4.6.1 Launcher brake test.- Erect launcher and install one mast section. Test for holding capability of the launcher brake by applying an axial mast load of 1000 pounds minimum to the raised mast section. Brake shall hold without slippage. For this test, mast shall not extend more than 50 inches above launcher. (See 3.9.1)

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4.6.2 Launcher bending test.- With the elevator at the upper or operational position and with the brake disengaged, a bending moment of 50,000 inch pounds minimum shall be applied to the launcher through an erected mast section individually in each of three directions. A bending moment shall be applied: (1) to produce compression in the rear launcher panel; (2) to produce tension in the rear launcher panel; and (3) to produce tensile and compressive forces in the two side launcher panels. The launcher shall be supported as in operation and shall meet the requirements of 3.9.2 after testing.

4.7 Winch tests.-

4.7.1 Load test.- An axial mast load of 500 pounds minimum shall be raised high enough to install the next section using the winch and the requirements of 3.10.1 shall be met.

4.7.2 Brake test.- The mast shall be lowered approximately 2 feet with the 500 pound minimum load using the winch brake only. The requirements of 3.10.2 shall be met.

4.7.3 Torque test.- The handle torque required to raise a 500 pound minimum axial mast load as in 4.7.2 shall be measured and meet the requirements of 3.10.3.

4.8 Mast Section tests.-

4.8.1 Bending test.- The two mast sections shall be coupled and loaded as a simple beam with the coupling centered. The beam shall be loaded transversely to cause a bending moment of 50,000 inch pounds minimum at the center of the span. Fixtures shall have provision to distribute applied loads such that failure due to local buckling does not occur. To obtain the required bending moment, two loads shall be applied on 30 inch centers with two reactions on 106 inch centers. (see Fig. 4). Loads and reactions shall be balanced about the center line of the joint. Two sets of coupled tubes shall be tested. Two sets of coupled tubes shall be tested, one set with the index tabs located on the neutral axis and one set with index tabs on the compression/tension axis. The mast sections shall meet the requirements of 3.11.1.

4.8.2 Bonding test.- The bending test of 4.8.1 shall be continued to destruction. The sections shall meet the requirements of 3.11.2.

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4.8.3 Indexing test.- The 50 foot mast assembled with mast cap shall be subjected to a 100 inch pound minimum torque. The angular displacement shall be measured and be in accordance with 3.11.3.

4.9 Guy tests.-

4.9.1 Guy assemblies and guy extension cables.- These items will be subject to a 2000 pound minimum tensile load and meet the requirements of 3.12.1.

4.9.2 Tensioner tests.-

4.9.2.1 Guy and tensioner assembly test.- Tensioners assembled to a guy assembly shall be subjected to a 2000 pound minimum tensile load with the safety latch disengaged. They shall meet the requirements of 3.12.2.1. Sample guys produced at the same time as the full length guys may be substituted in this test. These sample guys shall meet the requirements of 3.12.2.1. Failure of the sample guys shall be treated as failure of the full length guys.

4.9.2.2 Tensioner load test.- A 4000 pound minimum tensile load will be applied to the tensioner through the two 1/8 inch diameter cables extending from the jaws by clamping the cables together to share the load. After testing, the tensioner shall meet the requirements of 3.12.2.2.

4.9.2.3 Tension locking handle test.- A 40 pound minimum load shall be applied to the handle in the unlocking direction and perpendicular to the radius arm. After testing, it shall meet the requirements of 3.12.2.3.

4.10 Mast tests.-

4.10.1 Rotational resistance test.- With anchors located 35 feet from the center and fully guyed, the 50 foot mast shall be loaded as required by 3.17. The mast shall be rotated a full 360° both clockwise and counter-clockwise. The breakaway and running torques required to move the mast with the elevator in the operational mode and the mast brake system locked open shall meet the requirements of 3.17.

4.10.2 Loading tests.- (See 3.15)

4.10.2.1 Sustained loading test.- The 50 foot mast shall be erected as specified in 3.15.1 with a 60 pound minimum dummy antenna load mounted at the top. Horizontal loads equivalent to 90 mph winds

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are applied as required by 3.15.1 to the mast and antenna. The loads shall be directed to bisect the angle between adjacent guys. Provision shall be made at the load attachment points along the mast and at the dummy antenna which will permit mast rotation under load with a minimum of friction. All guys shall be initially loaded to approximately 100 pounds at all levels. There shall be no evidence of permanent deformation or failure under the conditions specified. The anchor radius for this test shall be 35 feet.

4.10.2.2 Erection load test.- With only the launcher guyed at 35 foot anchor radius, extend the mast to the 50 foot height while raising a 60 pound minimum dummy antenna and two 13 pound minimum dummy RF cable loads, one each attached to the top and mid guy rings. The top and mid guys shall be attached and left dangling. The mast brake shall be locked open and the 33 mph wind loads as required by 3.15.2 shall be applied. Mast shall meet the requirements of 3.15.2 after testing.

4.11 Bounce, loose cargo test.- (See 3.14.1) The launcher and mast sections with carrier prepared for transit, shall be tested per Method 514, Procedure XI, Part 2 of MIL-STD-810 as follows:

a. The total horizontal motion (both directions) shall be limited to not more than 2 inches and not less than 1 inch by wooden fences at least as high as the equipment.

b. The package tester shall be operated in the vertical linear mode only.

c. The launcher and mast carrier with four mast sections shall be tested separately or if tested simultaneously, they should be fenced separately.

d. The launcher and mast carrier with four mast sections shall be bounced for 45 minutes in each of the four positions. The positions shall be: the launcher, while resting on each of three sides with direction of elevator travel horizontal and while resting on the base; the mast carrier, with four mast sections, while resting on each of four sides.

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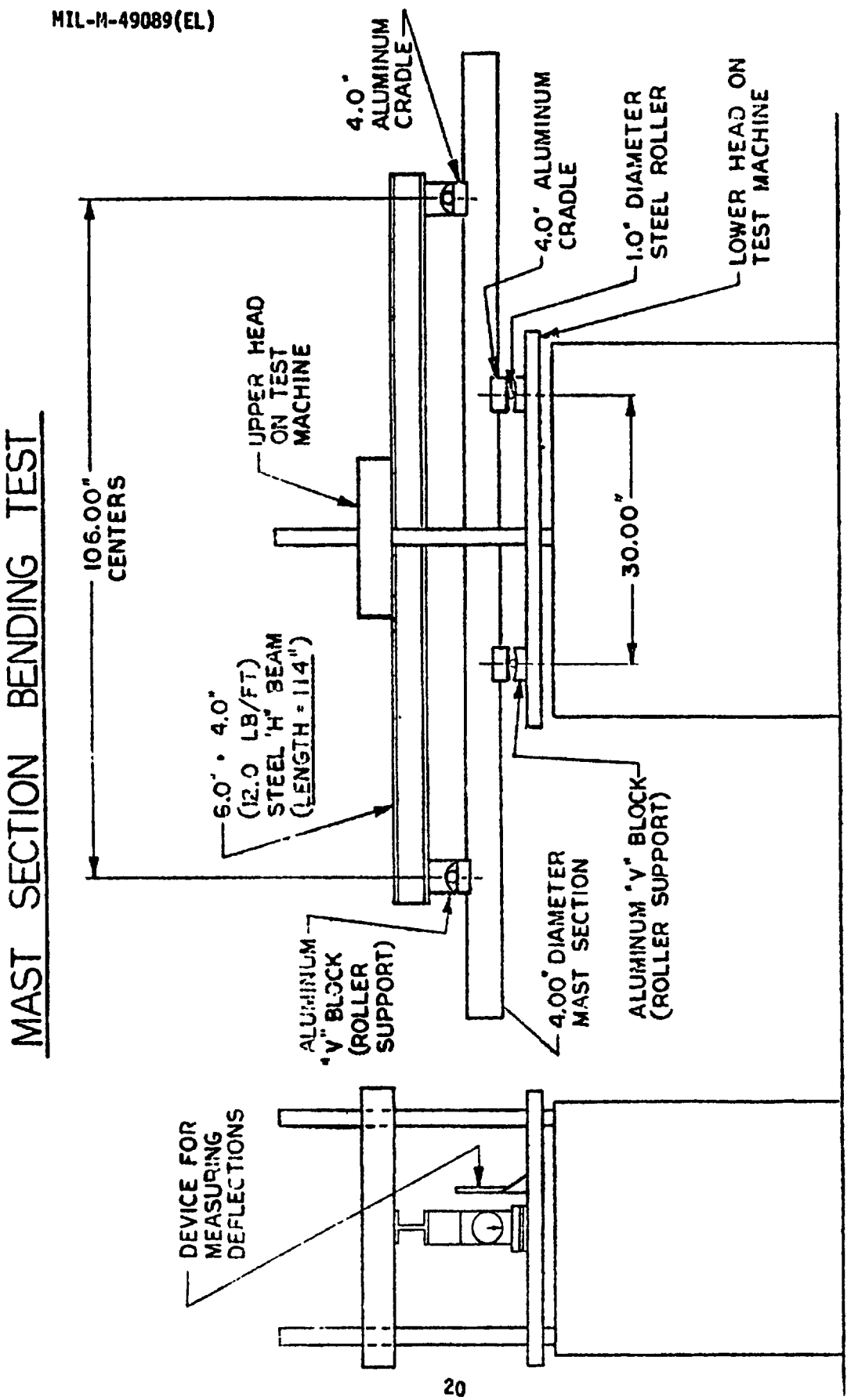


FIGURE 4

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4.12 Shock, drop test.- One launcher assembly and one mast section carrier with four mast sections, both assemblies in the transport mode, shall be subjected to Test Method 516, Procedure II of MIL-STD-810. The requirements of 3.14.2 shall be met.

4.13 Temperature tests.-

4.13.1 High temperature.- The complete mast in transit configuration shall be subjected to Test Method 501, Procedure II of MIL-STD-810. The high operating temperature for the equipment shall be 160°F. The requirements of 3.13 shall be met. (See 4.13.3)

4.13.2 Low temperature.- The complete mast in transit configuration shall be subjected to Test Method 502, Procedure I, of MIL-STD-810. The storage temperature shall be -70°F. The requirements of 3.13 shall be met. (See 4.13.3)

4.13.3 Measurements.- At Step 8, Test Method 501, Procedure II, and Step 5, Test Method 502, Procedure I, of MIL-STD-810, perform the tests of 4.19. The elevation mechanism, launcher brake and winch brake shall function smoothly with no binding, jamming or other operational defects. The mast section joints shall be operated at the same cycle stages and shall function smoothly. (See 4.13.1 and 4.13.2)

4.14 Humidity test.- The equipment shall be subjected to Test Method 507, Procedure I of MIL-STD-810 and meet the requirements of 3.13.

4.15 Altitude test.- The winch assembly shall be subjected to Test Method 500, Procedure I of MIL-STD-810 and meet the requirements of 3.13.

4.16 Dust test.- The launcher shall be subjected to Test Method 510, Procedure I of MIL-STD-810 and meet the requirements of 3.13.

4.17 Salt fog.- The equipment shall be subjected to Test Method 509, Procedure I of MIL-STD-810. Operation of the equipment shall not be required during the test and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish or metal surfaces. The requirements of 3.13 shall be met.

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4.18 Fungus test.- The mast with all accessories shall be tested per Method 508, Procedure I of MIL-STD-810. Those items normally stored in bags shall be removed and exposed during test. Operation is not required. The mast shall meet the requirements of 3.8.6.

4.19 Mast Operation Inspection.- The following operations shall be performed where specified, and meet the requirements of 3.16 to determine the results of other tests. (See 4.10, 4.11, 4.12, 4.13, 4.14, 4.15 and 4.16)

a. Operate lock pins on elevator, launcher lower section, base plate and locking screw for azimuth ring.

b. Mount winch on launcher.

c. Position one mast section in launcher and operate elevator.

d. Operate brake with mast section in place.

e. Install middle and upper guy rings and mast cap.

f. Rotate azimuth (with mast section) and each guy ring independently.

g. On a tensioner, operate the clamp handle and lock, and pull length of cable through the tensioner. Any malfunction or deterioration in ease of operation shall be cause for rejection (See 3.12.2) under the test being analyzed.

4.20 Inspection for interchangeability.- The dimensions listed below shall be gauged or measured to determine conformance to the physical interchangeability requirement of 3.18. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. Dimensions and location of torsional transfer elements.

b. Mating dimensions of section end fittings.

c. Mounting dimensions of separable assemblies.

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4.21 Visual and mechanical inspection.- The equipment with accessories shall be examined for the requirements of workmanship per 3.19 and the defects listed in Table V.

TABLE V

CLASSIFICATION OF VISUAL AND MECHANICAL DEFECTS

Classification	Defect
MAJOR	1. Welds: cracks, porosity, undercuts, gaps, burn-thru or distortion.
	2. Rivets: loose or heads not fully formed.
	3. Threaded fasteners: loose.
	4. Threads: cross-threaded, burred or mutilated.
	5. Locking screw and locking pins inoperable.
	6. Bags: buckles and snaps improperly fitted.
	7. Azimuth ring or pointers not properly installed.
	8. Winch mounting clips inoperable.
	9. Launcher rollers not freely rotating.
	10. Indexing marks and holes improperly located.
	11. Indexing pin inoperable.
	12. Upper guy ring not freely rotating.
	13. Dry film lubricant missing from mast cap, middle guy ring or mast section grooves.
	14. Rubber bumpers cracked or improperly attached.
	15. Nameplate, installation or brake caution plates incorrect or illegible.
MINOR	1. Finish: abrasions, scratches or flakiness.
	2. Guys not properly color coded.
	3. Cloth labels incorrect or illegible.
	4. Equipment not free of foreign material, rough finsishes.

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4.22 Quality conformance inspection of preparation for delivery.- (See 4.5) Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

4.23 Rough handling test (preparation for delivery).- When the rough handling test, in accordance with MIL-P-116, is required by the contract, the operational inspection of 4.19 shall be conducted to determine freedom from operational malfunction caused by rough handling.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified. (See 6.2b)

5.1.1 Level A.-

5.1.1.1 Cleaning.- Mast AB-864( )/G and Mast Extension Kit MK-1624( )/G shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying.- Mast AB-864( )/G and Mast Extension Kit MK-1624 ( )/G shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application.- None required.

5.1.1.4 Unit packaging.- Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature.- Each technical literature shall be packaged per Method 1C-1.

5.1.1.4.2 Mast AB-864( )/G.-

5.1.1.4.2.1 Mast sections.- Each mast section shall be individually packaged per Method III as follows: place the mast sections and components within the designated spaces of the mast carrier and secure with fasteners provided. No further packaging required.

5.1.1.4.2.2 Launcher.- No packaging required.

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5.1.1.4.2.3 Accessory bag and anchor roll with contents.- Each accessory bag and anchor roll with contents shall be individually packaged per Method III as follows: place the stakes, anchors, crowbar and hammer within the anchor roll with straps provided. Place the accessories and guys within the accessory bag and secure the bag with fasteners provided. Place both the accessory bag and anchor roll within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Fill all voids with paperboard conforming to PPP-P-291, type I, to prevent movement. Close the box in accordance with the appendix of the box specification.

5.1.1.4.3 Mast Extension Kit MK-1624( )/G.- Each mast extension kit shall be individually packaged per Method III in accordance with the procedures as specified in 5.1.1.4.2.1.

5.1.2 Level C.- Mast AB-864( )/G and Mast Extension Kit MK-1624 ( )/G shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing.- Packing shall be level A, B or C as specified. Shipping contrainers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container or its contents. (See 6.2.b)

5.2.1 Level A.- Each Mast AB-864( )/G and each Mast Extension Kit MK-1624( )/G, packaged as specified in 5.1, shall be individually packed within close-fitting boxes conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. Fiberboard boxes being placed directly in the shipping container shall be waterproofed with tape conforming to PP-T-76, in accordance with the taping requirements of the appendix of the box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, finish A.

5.2.2 Level B.- Each Mast AB-864( )/G and each Mast Extension Kit MK-1624( )/G, packaged as specified in 5.1, shall be individually packed within close-fitting boxes conforming to PPP-B-601, domestic

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type; PPP-B-621, style 4, class 1; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto.

5.2.3 Level C.- Mast AB-864( )/G and Mast Extension Kit MK-1624( )/G packaged as specified in 5.1, shall be individually packed in shipping containers that comply with Federal or Military specification in a manner that will afford adequate protection the package and its contents against physical and environmental deterioration and damage during limited shipment, handling and intransit storage.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- Mast AB-864( )/G and Mast Extension Kit MK-1624( )/G is used for support of Radio Antennas and associated transmission lines.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Levels of preservation and packaging and packing. (See Section 5)

c. First Article inspection: Three sample units of Mast AB-864( )/G and Mast Extension Kit MK-1624( )/G are generally required so that lengthy environmental tests can be completed on two sample units while complete performance measurements can be made on the third sample. (See 4.3)

d. Marking and shipping of samples.

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e. Place of final inspection.

f. Technical literature required.

g. Quantity of tools and running spare parts required.

h. Submission of the statement of treatment referenced in 3.8.5 as soon as possible after award of contract.

i. Data submissions per MIL-P-11268 require appropriate data items in the contract.

j. When rough handling test is required. (See 4.23)

6.3 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example, AB-864A/G or MK-1624A/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.4 Color of textiles.- Standard samples of olive drab No. 7 may be obtained from the Sample Load Unit, Defense Clothing and Textile Supply Center, Philadelphia, Pennsylvania 19101. When requesting samples, indication should be given as to type of material concerned.

6.5 Color.- The color chip No. 24087 may be obtained upon request to Commanding Officer, U.S. Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120, or as indicated in FED-STD-595.

6.6 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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Project No. 5985-A441

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-M-49089(EL) MAST AB-864( )/G And MAST EXTENSION KIT MK-1624( )/G

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes" in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-FM 1068-68

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